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Optoelectronic Synaptic Transistors Based on Metal-Oxide Semiconductors for Neuromorphic Vision Systems

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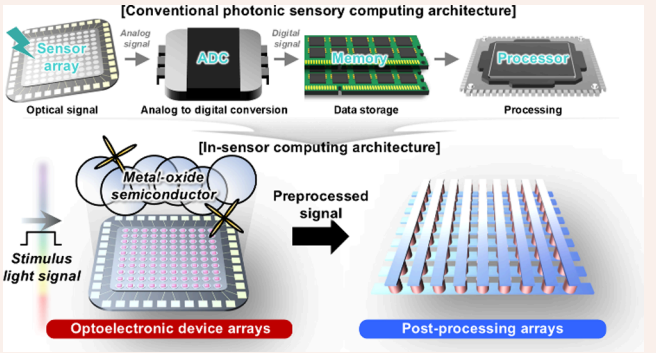
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CONSPECTUS: Metal-oxide semiconductor-based three-terminal devices are emerging as versatile building blocks for neuromorphic vision systems. Their unique appeal lies in the phenomenon of persistent photoconductivity (PPC), in which photocurrent persists after illumination is removed due to oxygen vacancy (V_O) mediated carrier trapping. This intrinsic memory-like behavior enables temporal integration, short-term retention, and synaptic plasticity within a single device, eliminating the need for external memory elements. Importantly, the presence of a gate electrode in three-terminal phototransistors provides a means to externally tune carrier generation and recombination, thereby regulating the PPC magnitude and lifetime. Such gate-dependent dynamics allow direct emulation of key synaptic functionalities including excitatory postsynaptic current (EPSC), paired-pulse facilitation (PPF), short-term to long-term plasticity (STP→LTP), and spike-timing-dependent plasticity (STDP). These characteristics establish metal-oxide semiconductor-based optoelectronic synaptic phototransistors as promising hardware platforms for in-sensor neuromorphic computing.



Despite these advantages, a critical limitation remains: pristine metal-oxide semiconductors possess wide bandgaps (typically >3 eV), confining photoresponse largely to the ultraviolet regime. For neuromorphic vision system applications, where broadband detection under visible and near-infrared (NIR) illumination is essential, this intrinsic spectral bottleneck restricts functionality. Addressing this challenge has therefore become a central research focus, and two complementary strategies have been developed to extend the photoresponse range while simultaneously exploiting PPC for synaptic operation.

The first strategy is trap-assisted photoexcitation, which relies on the intentional creation or activation of defect states within the metal-oxide semiconductor channel or at its interfaces. Treatments such as plasma exposure, chemical doping, or interfacial organic modification generate subgap states that act as intermediate energy levels. These trap sites enable absorption of lower-energy photons and enhance PPC by prolonging the carrier lifetimes. Through this mechanism, visible-light sensitivity is achieved, and synaptic functionalities, such as optical potentiation and electrical depression, are realized in a single device.

The second strategy is absorption layer engineering, which introduces external low-band gap materials onto oxide channels to provide broadband optical absorption. Hybrid integration with low-band gap materials facilitates carrier transfer into the metal-oxide semiconductor channel, where photogenerated electrons contribute to conductivity while holes are blocked or trapped at interfaces. This architecture not only broadens spectral sensitivity from UV into the visible and NIR regions but also sustains PPC through interfacial trapping, enabling tunable synaptic plasticity. In addition, material choice allows precise control of the absorption wavelength and persistence characteristics, thereby tailoring device performance to specific applications.

By leveraging defect engineering and heterojunction design, metal-oxide semiconductor-based optoelectronic synaptic transistors now perform core vision preprocessing tasks, such as contrast enhancement, adaptation, edge detection, and noise-tolerant image recognition directly at the sensor level. Looking ahead, the chemical robustness and low-temperature processability of metal-oxide semiconductors, combined with the compatibility for monolithic three-dimensional (M3D) integration, position these devices as scalable platforms for retinocortical in-sensor architectures. Continued advances in defect-state control, interfacial stability, and large-area fabrication will further accelerate the transition from laboratory demonstrations to high-density neuromorphic vision systems.

1. INTRODUCTION

The accurate and reliable perception of the physical world increasingly relies on the widespread deployment of vision-sensing nodes.[1,2] As the number of such devices continues to rise, the volume of visual data generated is growing at an unprecedented pace, creating tremendous opportunities while

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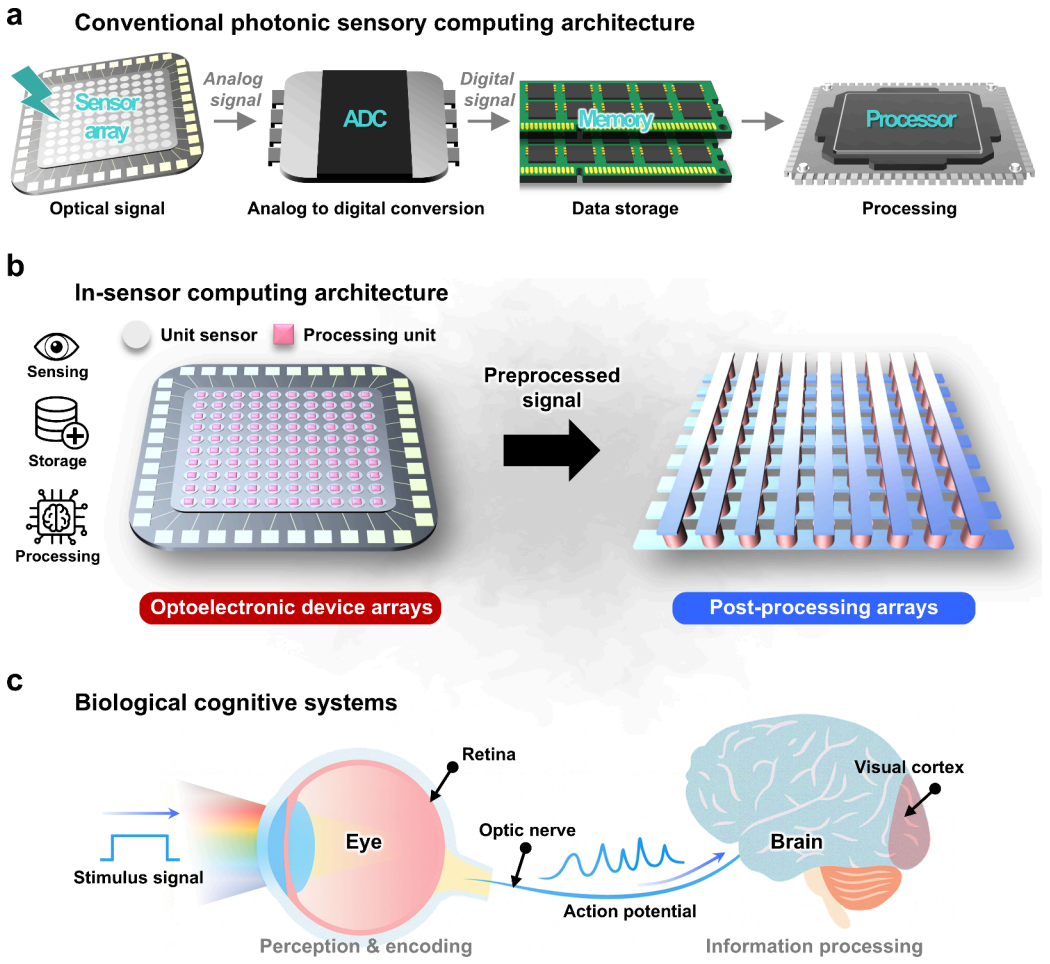


Figure 1. Imaging and visual data processing systems. (a) Schematic of a conventional sensory architecture, where external stimuli are captured by sensor arrays, converted into digital signals via analog-to-digital converters, stored in memory, and subsequently processed by digital processors. (b) Neuromorphic vision system based on in-sensor computing, employing multifunctional devices that serve concurrently as photosensors and artificial synapses within the front-end processing stage. (c) Biological visual recognition system consisting of the eye and brain. Incident light is focused onto the retina by a single lens, where photoreceptor cells are activated and the encoded visual signals are transmitted to the visual cortex as action potentials.

simultaneously imposing serious challenges for data processing and system efficiency.[3] In conventional von Neumann–based computing systems, sensors simply deliver raw environmental data, which are then sequentially processed by discrete modules such as analogue-to-digital converters, memory units, and digital processors, as shown in Figure 1a.[4,5] However, this repeated transfer of raw data across physically separated components inherently limits operating speed and energy efficiency and further raises concerns regarding data privacy and security.[5] Moreover, the continual movement of large-scale image data not only leads to excessive energy consumption but also induces latency and bandwidth bottlenecks.[6]

From this perspective, the ubiquitous deployment of next-generation neuromorphic vision sensors capable of advanced image recognition and data preprocessing will accelerate progress in applications ranging from mobile devices and autonomous vehicles to healthcare monitoring and industrial automation.[7,8] Neuromorphic vision sensors are generally designed to be suitable for neuromorphic vision computing tasks such as noise removal, edge enhancement, spectral filtering, and visual information recognition, and are emerging first in the paradigm of near-sensor computing and subsequently in in-sensor computing.[9−11] In particular, in-sensor computing integrates perception, memory, and processing functions within the sensor itself, enabling sensor terminals to extract essential information and perform data compression and structuring. When information is processed directly in the sensor terminals, the preprocessed data transmitted to subsequent processing units, such as synaptic memristor arrays, is reduced as shown in Figure 1b.[1,12,13] In other words, by processing information at the point of acquisition in real time, the volume of data transmitted to subsequent units can be effectively reduced.

Achieving ultimate in-sensor computing devices requires significant advances in materials science, device engineering, and computing architectures. A representative strategy lies in the development of neuromorphic devices inspired by the biological visual system.[14] In the human eye, the retina not only generates high-resolution visual images but also performs critical preprocessing operations such as noise suppression and image sharpening (Figure 1c).[14,15] These functions reduce redundant information and mitigate transmission-induced delays, thereby lowering energy consumption and enhancing efficiency. Inspired by this paradigm, artificial vision systems aim to replicate such retinal capabilities, providing temporary

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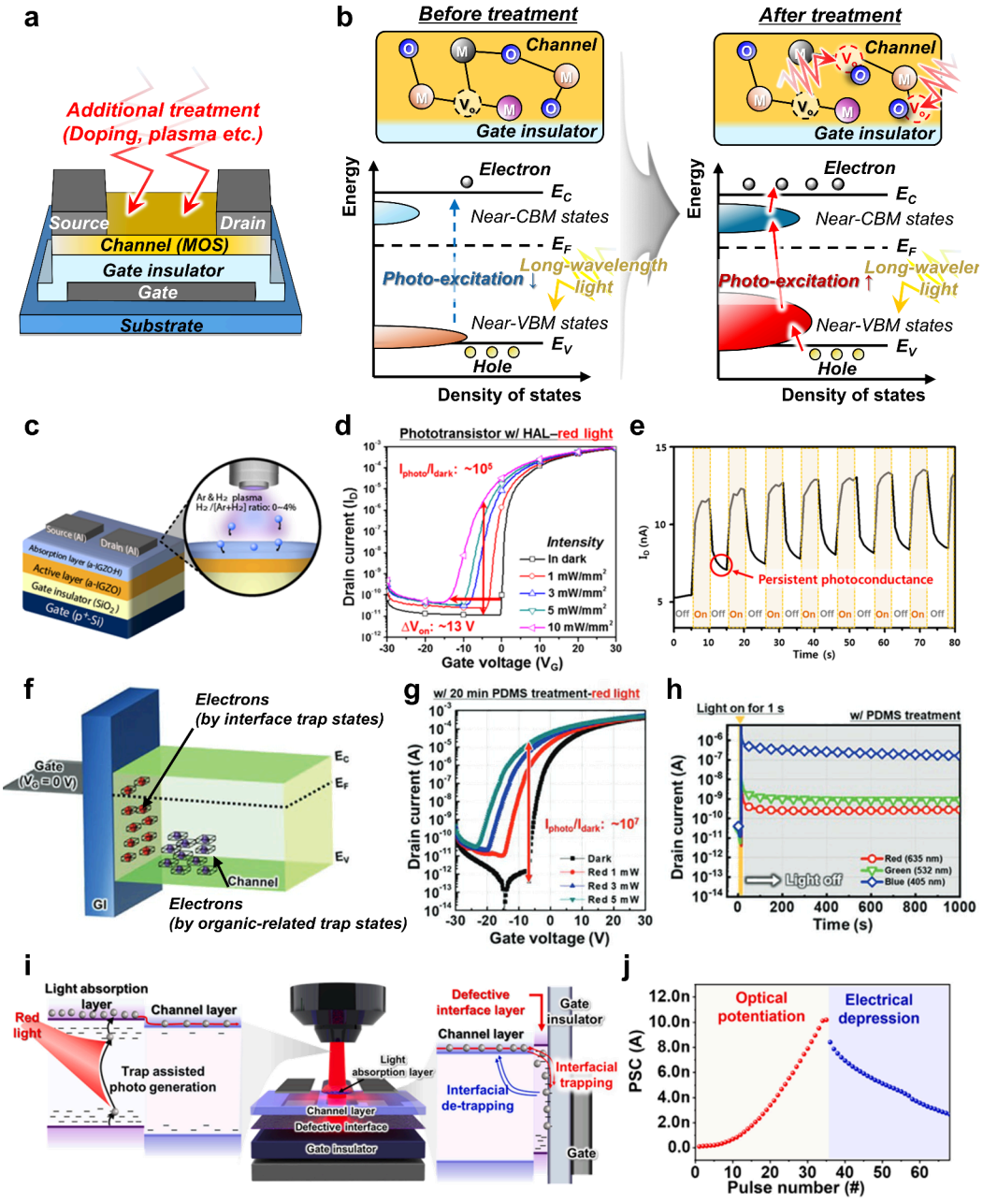


Figure 2. Trap-assisted photoexcitation in metal-oxide semiconductor-based optoelectronic synaptic transistors. (a) Schematic illustration of channel modification (doping, plasma treatment, etc.) for generating trap states in metal-oxide semiconductors. (b) Creation of oxygen-vacancy-related trap sites via bond breaking under photoexcitation, enabling subgap photon absorption and extended wavelength response. (c) Device structure of an IGZO phototransistor with hydrogen/argon plasma treatment. (d) Transfer characteristics under red light illumination (635 nm). (e) Repetitive pulse measurements demonstrating PPC with a cumulative photocurrent increase. Reproduced with permission from ref 39. Copyright 2018 American Chemical Society. (f) Schematic of interface and organic-related trap generation in IGZO phototransistors. (g) Transfer curves of IGZO devices after PDMS treatment under red light illumination (635 nm). (h) Time-dependent photocurrent measurements confirming PPC across the visible wavelengths. Reproduced with permission from ref 40. Copyright 2022 Wiley-VCH. (i) Multilayer channel engineering by stacking solution IGZO films. (j) Synaptic functionalities derived from trap-assisted mechanisms, including optical potentiation and electrical depression. Reproduced with permission from ref 41. Copyright 2023 Elsevier.

memory responses to optical stimuli through simplified device structures, while offering promising solutions to the inefficiencies of conventional architectures.

To enable these functions in hardware, extensive research has been conducted on two-terminal and three-terminal optoelectronic devices based on a wide variety of semiconductor materials.[16,17] By engineering light−matter interactions, optoelectronic devices can simultaneously sense, memorize, and process optical information, mapping visual inputs into compressed or structured representations. Among the physical mechanisms available, persistent photoconductivity (PPC) has emerged as one of the most powerful enablers of neuromorphic vision hardware.[1,9,14,18] PPC originates from the trapping of photogenerated carriers within localized defect states, resulting in prolonged conductivity even after the light stimulus is removed.[19] This property provides intrinsic

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memory-like retention of optical information and allows devices to integrate temporal signals across multiple input events. In practice, PPC enables optoelectronic devices to perform short-term memory operations, synaptic plasticity, and adaptive learning-like behaviors.[7] These functionalities are essential for processing dynamic visual scenes in real time without relying on external memory elements. Another critical point is the presence of a gate terminal that can modulate PPC after light stimulation by influencing carrier generation and recombination.[20,21] While two-terminal devices such as photodiodes or memristors have demonstrated PPC-based synaptic characteristics, their controllability is inherently limited. In contrast, three-terminal devices, such as phototransistors, provide an additional gate electrode that allows external modulation of PPC behavior.[21] This tunability is particularly advantageous for achieving large dynamic ranges, noise suppression, and reconfigurable operation modes.[22] As such, three-terminal architectures are widely regarded as superior candidates for implementing robust sensor computing functions. In this context, among the various material systems investigated—including organic semiconductors, two-dimensional materials, and perovskites—three-terminal devices based on metal-oxide semiconductors, which inherently possess PPC characteristics, can be regarded as promising channel materials and device structures for ultimately realizing in-sensor computing.[9,20,23,24] In addition, metal-oxide semiconductors offer a synergistic configuration of electrical performance, chemical robustness, and compatibility with low-temperature and scalable fabrication processes.[25−29] These properties make them attractive as both channel materials and integration platforms for large-area vision arrays and monolithic three-dimensional (M3D) integration.

In this Account, we provide a comprehensive overview of the hardware implementation of three-terminal, metal-oxide semiconductor-based optoelectronic devices for in-sensor computing. We discuss the underlying mechanisms of photogenerated carrier dynamics and the intrinsic limitations of metal-oxide semiconductors, along with the biological principles and neural functionalities that artificial vision systems seek to emulate, in the Supporting Information. First, we examine device structures and operation principles that overcome intrinsic material limitations when employed as three-terminal optoelectronic devices and exploit PPC characteristics. This is followed by a discussion of array-level demonstrations and application studies that showcase in-sensor computing in practice. Finally, we propose future research directions and highlight the opportunities and challenges for realizing ultimate neuromorphic vision systems based on metal-oxide semiconductors.

2. MECHANISMS FOR EXTENDING SPECTRAL SENSITIVITY AND EXPLOITING PPC IN METAL-OXIDE SEMICONDUCTOR-BASED OPTOELECTRONIC SYNAPTIC TRANSISTORS

To streamline the discussion, a comprehensive overview of the basic functionalities of metal-oxide semiconductor-based optoelectronic synaptic transistors, together with the underlying oxygen-vacancy ionization/neutralization mechanisms as shown in Figure S1 and key synaptic parameters as shown in Figure S2.[9,30−36,53,54]

In the application of metal-oxide semiconductors to optoelectronic synaptic transistors, one of the most critical challenges stems from their intrinsically wide bandgap, which restricts photoresponse to the ultraviolet or blue spectral range. Such narrow spectral sensitivity significantly limits their potential for neuromorphic vision systems, where broadband detection and dynamic plasticity under diverse optical stimuli are required. To overcome this limitation, researchers pursued two representative strategies that simultaneously expand the detection wavelength and exploit PPC for synaptic functionalities. The first approach relies on trap-assisted photoexcitation, in which intentional introduction of defect states enables subgap photon absorption and enhances PPC through defect-mediated carrier trapping.[37−41] The second approach employs additional low-bandgap absorption layers integrated with the metal-oxide semiconductor channel, thereby extending the spectral response while maintaining transistor operation.[42−46] Together, these mechanisms define the primary research directions for addressing the intrinsic bandgap bottleneck and advancing oxide semiconductors toward practical optoelectronic synaptic devices.

2.1. Trap-Assisted Photoexcitation Engineering for Wavelength Expansion and Syanptic Plasticity

Trap-assisted photoexcitation has emerged as one of the most effective strategies to overcome the wide bandgap limitation of metal-oxide semiconductors and extend their spectral sensitivity for optoelectronic synaptic applications. The intrinsic electronic structure of metal-oxides confines absorption largely to the ultraviolet regime, which is insufficient for practical neuromorphic vision systems that must operate under broadband illumination conditions. By intentionally introducing trap states within the metal-oxide semiconductor channel or at its interfaces, subgap photons can be absorbed through defect-mediated transitions, thereby enabling a photoresponse at longer wavelengths. Moreover, these defect sites naturally enhance PPC, which is a key property for mimicking synaptic plasticity.

A deeper understanding of this mechanism is illustrated in Figure 2a and b. In pristine metal-oxide semiconductors, photon absorption occurs primarily through direct band-to-band transitions, requiring photon energies greater than the intrinsic bandgap. However, treatments such as doping, plasma exposure, or chemical modification, including interface coupling and compositional engineering, can partially disrupt the strong metal−oxygen bonds, generating V_O-related trap states located near the conduction- and valence-band edges.[39−41,55−57] Similarly, metal-oxide semiconductor films deposited under oxygen-deficient atmospheres inherently possess a high density of V_O, which likewise introduce subgap states and extend the optical response into the visible wavelength range. These trap sites serve as intermediate energy levels, enabling subgap photons to excite electrons either from the valence band into a trap state and subsequently into the conduction band or directly from the trap state into the conduction band. This excitation pathway effectively lowers the absorption threshold and broadens the spectral response to the wavelength above UV.

One representative demonstration of this approach is shown in Figure 2c−e, where Ar/H_2 plasma treatment was applied to IGZO channels.[39] Plasma exposure effectively increased the density of oxygen vacancies and introduced hydrogen-related bonding configurations. As a result, devices exhibited sensitivity to red light at 635 nm, far below the absorption edge of pristine IGZO (Figure 2d). The treated devices also

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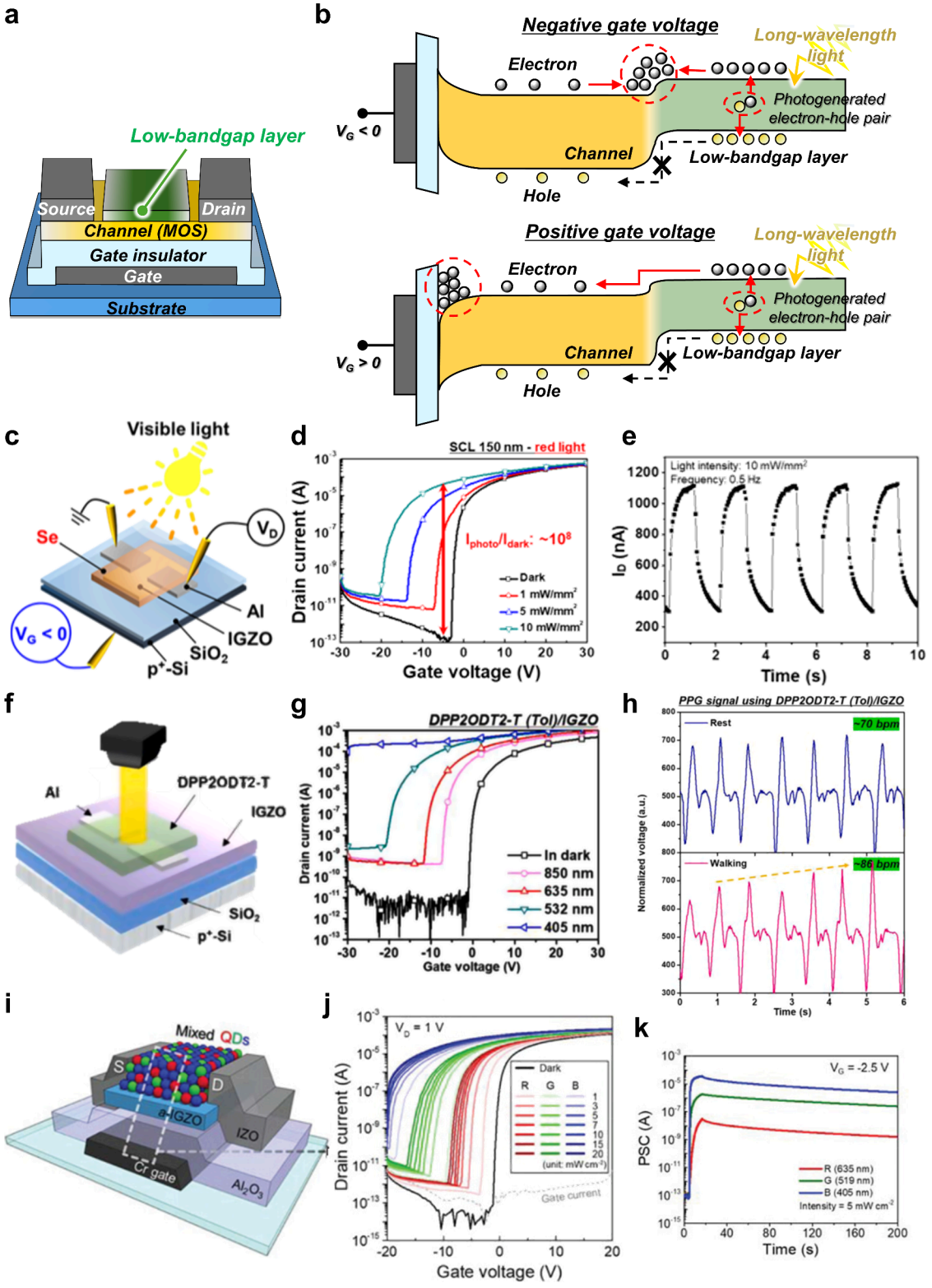


Figure 3. Absorption layer engineering for broadband photoresponse and synaptic plasticity. (a) Device schematic of a metal-oxide channel integrated with a low-bandgap absorption layer. (b) Band diagram illustrating bias-dependent carrier transfer from the absorption layer to the metal-oxide semiconductor channel under long-wavelength illumination. (c) Selenium (Se)/IGZO hybrid phototransistor structure for extended visible-light detection. (d) Transfer characteristics of Se/IGZO hybrids under red light (635 nm). (e) Time-resolved photocurrent of Se/IGZO devices under a pulsed red light. Reproduced with permission from ref 42. Copyright 2020 American Chemical Society. (f) Hybrid device with conjugated polymer DPP2ODT2-T stacked on IGZO channel. (g) Spectral photoresponse of DPP2ODT2-T/IGZO hybrids from 405 to 850 nm, revealing broadband sensitivity. (h) Photoplethysmography (PPG) signal detection using polymer/IGZO hybrids, capturing heart rate signals under resting and walking states. Reproduced with permission from ref 43. Copyright 2022 Elsevier. (i) Device schematic of a mixed quantum dot (QD)/IGZO heterojunction. (j) Transfer curves of QD/IGZO devices under RGB illumination, confirming broadband absorption and gate-

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Figure 3. continued

modulated photoresponse. (k) PPC characteristics of QD/IGZO hybrids at different wavelengths, demonstrating wavelength-dependent synaptic potentiation. Reproduced with permission from ref 44. Copyright 2022 Wiley-VCH.

displayed dramatically enhanced PPC, as shown in Figure 2e, with photocurrent levels steadily accumulating under repeated optical pulses. The PPC in these devices mainly derives from defect states generated within the hydrogen-doped absorption layer (HAL) and at the interface between the active layer and the HAL, which act as reservoirs for long-lived carriers. This cumulative response is highly analogous to synaptic potentiation in biological systems, where successive spikes reinforce the synaptic weight. The ability to induce such learning-like behavior by tuning plasma parameters highlights the flexibility of trap-assisted strategies in controlling device functionalities.

Another approach involves polydimethylsiloxane (PDMS) treatment at the IGZO/SiO_2 interface, which introduces interfacial and organic-related trap states.[40] Figure 2f–h demonstrates an example in which PDMS treatment was applied at the IGZO/SiO_2 interface. This organic modification introduced both interfacial and organic-related trap states that strongly couple to the IGZO channel. As a result, the devices achieved pronounced subgap photoresponse to visible light, as shown in Figure 2g. Moreover, these devices exhibited PPC across multiple visible wavelengths (Figure 2h), with residual currents persisting even after prolonged optical stimulation. Such behavior underscores the importance of interface engineering not only for spectral broadening but also for enabling memory retention.

As another strategy, researchers have stacked solution-processed IGZO layers above and below a primary IGZO channel, as illustrated in Figure 2i.[41] This configuration led to the formation of abundant interface defect states as well as organic-related defect states within the IGZO channel layer. Such multilayer structures facilitate efficient absorption of longer-wavelength light and dramatically prolonged carrier lifetimes. The neuromorphic relevance of this architecture is evident in Figure 2j, where the device exhibited clear bidirectional synaptic behaviors. Specifically, optical pulses potentiated the channel conductance via trap-assisted excitation, while electrical signals depressed the conductance by facilitating the recombination and neutralization of ionized vacancies. This dual-mode control mimics the potentiation and depression processes of biological synapses, demonstrating the potential of trap-engineered multilayer oxides for artificial learning systems.

An alternative route involves metal-oxide semiconductor-based optoelectronic devices fabricated under oxygen-deficient conditions where a high density of V_O introduces subgap states that enable visible-light absorption. Notably, Zhuge et al. demonstrated an all-optically controlled IGZO-based devices fabricated under oxygen-poor conditions, in which vacancy-mediated trap states enabled wavelength-selective optical potentiation and depression, resulting in nonvolatile and bidirectional synaptic plasticity.[53] Similarly, Ye et al. reported ZnO-based devices sputtered in a pure Ar atmosphere, where abundant V_O extended the photoresponse into the visible range and enabled optical long-term potentiation as well as depression via trap-assisted carrier ionization and neutralization processes.[54]

Taken together, these diverse examples highlight the versatility of trap-assisted photoexcitation as a unifying design principle for overcoming the intrinsic bandgap bottleneck in metal-oxide semiconductors. Plasma treatments generate oxygen-vacancy-rich environments that extend sensitivity to subgap photons and enhance PPC. Interface engineering introduces tailored traps at critical junctions to achieve both broad-band detection and long-term memory behavior. Multilayer stacking creates dense trap networks that enable robust and tunable synaptic plasticity, including optical potentiation and electrical depression. Oxygen-deficient deposition conditions inherently generate vacancy-rich metal-oxide semiconductors, introducing subgap states that extend the photoresponse into the visible range and enable PPC–driven long-term plasticity (LTP). Each of these approaches leverages the same underlying mechanism—defect-assisted absorption and prolonged carrier lifetimes—yet offers distinct pathways for tailoring device performance.

In summary, trap-assisted photoexcitation provides a versatile and material-compatible approach to broaden the spectral sensitivity and amplify PPC in oxide semiconductors. By carefully designing defect structures, researchers have succeeded in extending the photoresponse from the ultraviolet into the visible and red-light regimes while simultaneously reinforcing synaptic functionalities. These advancements confirm that what might traditionally be considered a material drawback—namely, the presence of defects—can instead be harnessed as a powerful enabler of neuromorphic operation. As such, trap-assisted strategies are likely to remain a cornerstone for the continued development of metal-oxide semiconductor-based optoelectronic synaptic transistors and their integration into in-sensor neuromorphic vision platforms.

2.2. Absorption Layer Engineering for Broadband Photoresponse and Syanptic Plasticity

While trap-assisted strategies provide one pathway to broaden the spectral sensitivity of metal-oxide semiconductors, an alternative and complementary approach involves the introduction of additional low-bandgap absorption layers. This method overcomes the inherent bandgap limitation of metal-oxide semiconductors by incorporating materials that enable absorption of longer wavelength light beyond the UV and transport of generated carriers into the metal-oxide semiconductor channel. Such hybrid structure phototransistors effectively combine the broadband absorption capabilities of chalcogen-based absorber, organic materials, halide perovskites, or quantum dots (QDs).[42−44,58] In addition to extending detection into the visible and near-infrared (NIR) regimes, absorption layers also contribute to PPC generation, thereby enhancing the synaptic functionalities of the device.

As shown in Figure 3a and b, a low-bandgap absorption layer is stacked directly on top of the metal-oxide semiconductor channel. In the hybrid architecture, a low-band gap absorption layer forms a heterojunction with the metal-oxide semiconductor channel, creating conduction- and valence-band offsets that assist carrier separation. Upon illumination, electron-−-hole pairs are generated in the absorption layer. The gate bias then governs band bending at the interface and,

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consequently, the preferred transfer pathway. With a negative gate voltage (channel depletion), the interfacial field facilitates electron extraction from the absorber into the n-type metal-oxide semiconductor while blocking holes. With a positive gate voltage (channel accumulation), the photogenerated electrons are transferred from the absorption layer to the front channel of the current to increase. As a result, more electrons flow into the front channel, causing the current increase. If the wavelength of illuminated light is shorter, then the energy of light is stronger, forming more photogenerated electrons. Depending on the applied gate bias, one type of carrier (electron or hole) is injected into the metal-oxide semiconductor channel, where it modulates the conductivity and contributes to the photocurrent. Electrons trapped at the interface between the absorption and channel layers or at ionized oxygen vacancies within the metal-oxide semiconductor give rise to long-lived responses that manifest as PPC. This architecture thus enables both broadband photoresponse and synaptic plasticity within the same device.

One representative example is shown in Figure 3c–e, where selenium (Se), with a bandgap of 1.95 eV, was deposited onto IGZO channels.[42] In this structure, photocarriers are generated in the Se layer under red-light illumination; electrons then transfer into the IGZO channel, while holes are blocked or trapped at the Se/IGZO interface, thereby reducing recombination. As shown in Figure 3d, the hybrid Se/IGZO transistor exhibited a dramatic increase in photocurrent under red illumination, with an I_{photo}/I_{dark} ratio approaching 10^9. Furthermore, the device displayed a strong and reproducible PPC response to pulsed red light stimuli, as demonstrated in Figure 3e. This sustained current decay is directly analogous to synaptic potentiation and memory behavior, establishing Se/IGZO hybrids as promising candidates for broadband optoelectronic synaptic transistors.

Organic materials represent another important class of absorption materials due to their tunable bandgaps and solution processability.[43] Figure 3f–h demonstrates a hybrid transistor in which the conjugated polymer diketopyrrolopyrrole (DPP) was integrated with an IGZO channel. In this architecture, photons absorbed by the DPP polymer layer generate excitons; these excitons dissociate to free carriers, and electrons are injected into the IGZO channel, while holes remain in the polymer or are trapped at the interface, thereby creating a photogating effect. Moreover, by employing different conjugated polymers (DPP polymers dissolved in chlorobenzene and toluene) with varied bandgaps and carrier dynamics, the magnitude and decay time of PPC can be systematically modulated, offering an additional degree of control over synaptic plasticity. The polymer layer broadened the spectral sensitivity of the device into visible and NIR regions, with photoresponse at 405, 532, 635, and even 850 nm (Figure 3g). Importantly, the hybrid device was employed for photoplethysmography (PPG) signal detection, as shown in Figure 3h, to capture physiological signals under different activity states (resting and walking). This example underscores the multifunctionality of absorption layer engineering, enabling both broadband vision-like detection and biomedical sensing applications, while maintaining synaptic behaviors through PPC-driven responses.

Inorganic nanostructures such as QDs also provide unique advantages as absorption layers.[44] Figure 3i presents a device where mixed red, green, and blue QDs were deposited onto an IGZO channel. Owing to their size-dependent bandgap tunability, QDs collectively enabled broadband absorption covering the visible spectrum. The hybrid QD/IGZO transistors exhibited strong gate-modulated photocurrents under RGB illumination (Figure 3j), with persistent responses maintained after light pulses were removed (Figure 3k). The synaptic relevance of these devices lies in their ability to mimic wavelength-dependent potentiation, where optical spikes of different colors induce distinct magnitudes of conductance modulation, akin to multisensory integration in biological systems.

Compared with trap-assisted photoexcitation, absorption layer engineering offers several complementary advantages. First, the spectral sensitivity can be precisely tuned by selecting appropriate low-bandgap materials, enabling detection well beyond the intrinsic band edge of metal-oxides. Second, hybrid integration allows the device to combine the high carrier mobility and thermal stability of metal-oxide semiconductors with the strong absorption of semiconductors, polymers, or QDs. Finally, absorption layers introduce additional pathways for PPC at heterojunction interfaces, thus contributing to synaptic plasticity.

Nevertheless, this approach also faces unique challenges. The fabrication of uniform and stable absorption layers requires careful control of the deposition processes to ensure reliable carrier transfer into the metal-oxide semiconductor channel. Organic layers may suffer from limited environmental stability, while quantum dots can introduce interfacial defects or variability due to the size distribution.

In summary, absorption layer engineering provides a robust strategy to extend the photoresponse of metal-oxide semiconductor-based optoelectronic synaptic transistors into the visible and NIR ranges. Through hybridization with low-band gap semiconductors, organic polymers, or QDs, devices exhibit tunable PPC, enabling memory-like behavior in response to a wide spectrum of optical stimuli. The diverse examples highlight the versatility of this approach, from Se/IGZO hybrids to broadband organic/oxide and QD/oxide heterojunctions. Together with trap-assisted photoexcitation, absorption layer integration establishes the second major pathway for overcoming the intrinsic bandgap limitations of oxide semiconductors, paving the way toward broadband, multifunctional optoelectronic synaptic devices for neuromorphic vision systems.

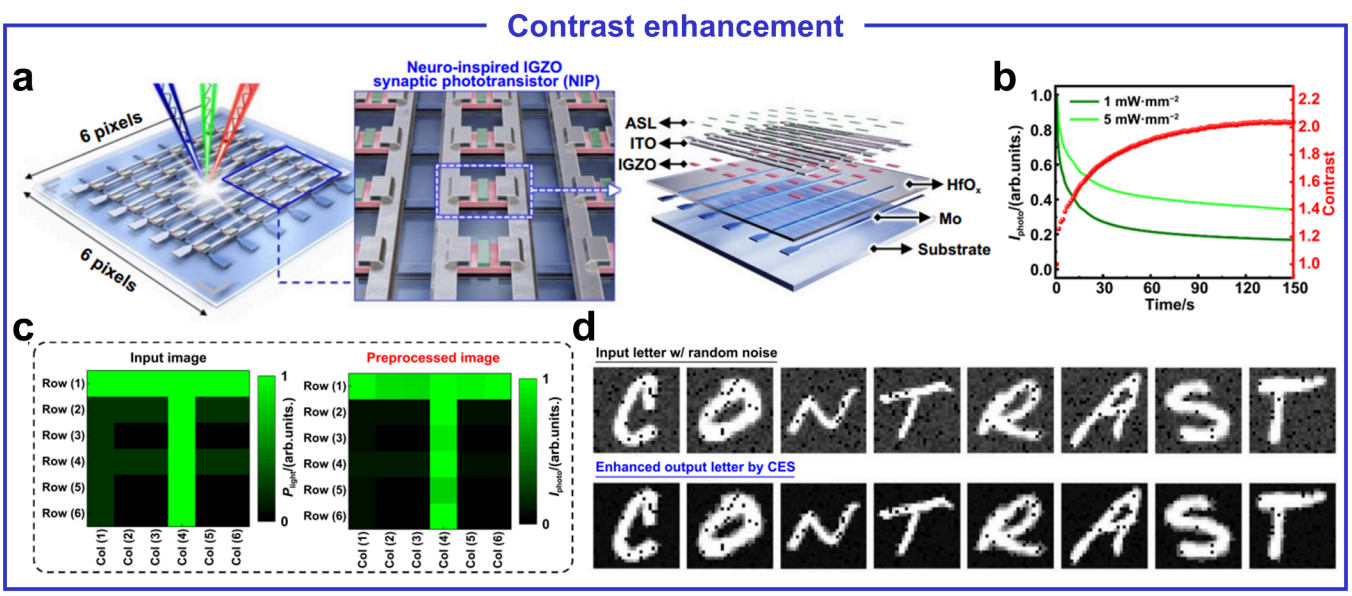
3. ADVANCED APPLICATIONS OF OPTOELECTRONIC SYNAPTIC TRANSISTORS FOR IN-SENSOR COMPUTING

Recently, researchers were seeking advanced approaches to extend the functionality of optoelectronic synaptic transistors beyond simple memory and plasticity, aiming to implement core vision tasks directly within the sensor hardware. By exploiting light-modulated conductance, PPC, and gate-tunable dynamics, these devices have been adapted to perform fundamental image preprocessing and recognition functions. Representative examples include in-sensor contrast enhancement, adaptation, edge detection, and image recognition, highlighting their potential as versatile building blocks for neuromorphic vision systems.

3.1. In-Sensor Contrast Enhancement

Recently, many efforts have been made to build neuromorphic vision sensors that can enhance the contrast of acquired images. In neuromorphic vision sensors, the effective

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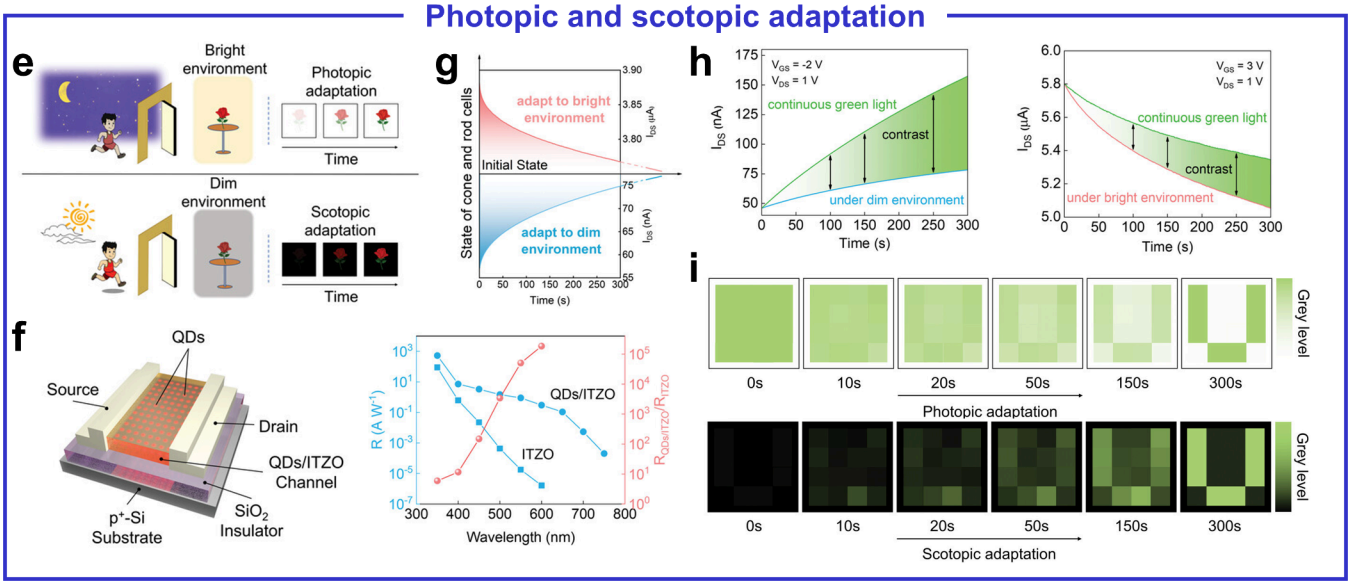


Figure 4. Optoelectronic synaptic transistors for in-sensor contrast enhancement and visual adaptation. (a) Schematic of a 6 × 6 IGZO synaptic phototransistor array with an Al-sensitization layer (ASL) for enhanced visible light absorption. (b) Photocurrent decay characteristics and contrast variation under light irradiation of 1 and 5 mW/mm^2, showing slower decay and higher residual current at higher intensity. (c) Grayscale distribution of input and preprocessed images illustrating contrast enhancement between target and background pixels. (d) Demonstration of robust letter recognition against random noise, highlighting the improvement in output image clarity. Reproduced with permission from ref 9. Copyright 2025 IOP Publishing Ltd. (e) Schematic illustration of photopic and scotopic adaptation in human vision. (f) Schematic illustration of InP QDs/ITZO hybrid phototransistors, with spectral absorption. (g) Different levels of remaining photocurrent over time, analogous to the states of photoreceptor cells. (h) Time-dependent current of the hybrid phototransistors under bright and dim environments. (i) Time-resolved images obtained from a 4 × 4 device array, demonstrating gradual contrast evolution under bright and dim environments, corresponding to photopic and scotopic adaptation, respectively. Reproduced with permission from ref 47. Copyright 2023 Wiley-VCH.

modulation of the conductivity relaxation rate is crucial as it depends on the intensity and temporal characteristics of the light stimulus. Such tunable relaxation dynamics allow the devices to selectively emphasize critical visual features while suppressing redundant background information, thereby enabling experimental implementation of image preprocessing for contrast enhancement.[3] In a pioneer study on contrast enhancement using optoelectronic synaptic transistors, Kim et al. reported capability of contrast enhancement using 6 × 6 neuro-inspired optical image sensor array for in-sensor computing, as shown in Figure 4a.[9] The optoelectronic synaptic transistor was fabricated using an IGZO channel and aluminum (Al)-sensitization layer (ASL). Utilization of ASL on IGZO not only enhanced the visible light absorption capabilities but also adjusted the PPC behavior. The functions of the retina, such as noise reduction and edge enhancement, improve the efficiency of the cortex in processing large volumes of visual data. Inspired by the preprocessing capabilities of the retina, the unique synaptic plasticity of these devices enables IGZO synaptic phototransistors to exhibit light-tunable conductance, facilitating various low-level sensory processing tasks. By leveraging the dependence on light dosage for photogeneration and the temporally resolved characteristics of historical inputs, this neuro-inspired

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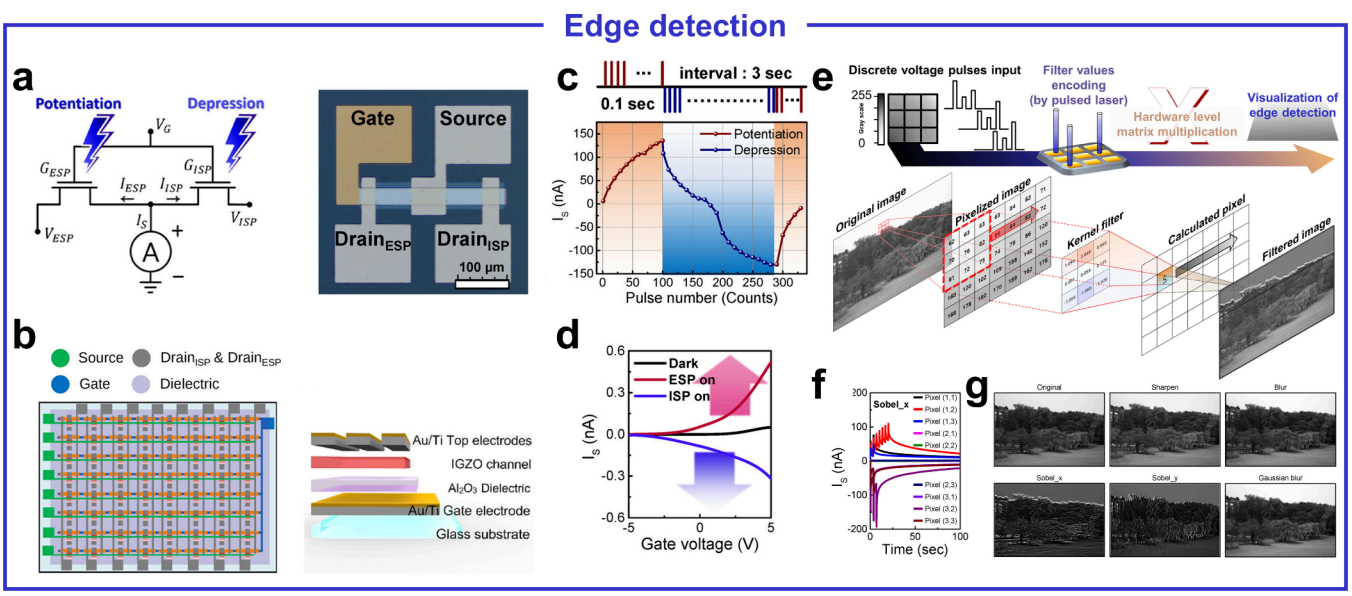


Figure 5. (a) Circuit configuration and optical images of serially connected IGZO-based excitatory synaptic phototransistor (ESP) and inhibitory synaptic phototransistor (ISP). (b) Schematic of device array architecture. (c) Synaptic current modulation under repeated optical pulses, showing potentiation and depression characteristics. (d) Variation in excitatory and inhibitory characteristics depending on different gate bias. (e) Schematic illustration of the image preprocessing workflow using convolution kernels programmed into synapse arrays. (f) Programmed synaptic responses corresponding to the Sobel_x kernel. (g) Processed images demonstrating sharpening, blurring, Sobel edge detection, and Gaussian blurring. Reproduced with permission from ref 50. Copyright 2023 Wiley-VCH. (h) Hybrid device architecture combining IGZO-based optoelectronic synaptic phototransistors with quantum dot light-emitting diodes (QLEDs) for noise-tolerant image recognition. (i) Comparison of current responses of IGZO-based optoelectronic synaptic phototransistors and QLED under optical pulse irradiation. (j) Experimental demonstration of noise filtering and improved image clarity. (k) System-level schematic of image preprocessing with integrated hardware- and software-based deep neural network classification. (l) Recognition accuracy comparison showing enhanced performance compared to noisy inputs. Reproduced with permission from ref 51. Copyright 2022 American Association for the Advancement of Science.

optical sensor integrates sensing, memory, and contrast enhancement for image processing. Figure 4b demonstrates the decay characteristics under 64 light pulses with intensities of 1 and 5 mW/mm^2. The higher intensity (5 mW/mm^2) resulted in a slower decay and a higher remaining photocurrent ratio, approximately twice as large after 150 s. This adjustable conductance relaxation speed aids in enhancing the contrast of input images by emphasizing critical features of the target letter against the noise. Figure 4c illustrates the grayscale distribution of input images featuring the target letter 'T', illuminated at 5 mW/mm^2, and the background letter 'F', illuminated at 1 mW/mm^2. The processed image, after repeated stimulation, displayed a marked improvement in contrast between the input pixels, yielding an output image with improved clarity and features. Figure 4d represents robustness against random white noise under varying light intensity and decay time conditions using Extended Modified National Institute of Standards and Technology (EMNIST) data set. The findings indicate that preprocessing with the neuro-inspired optical image sensor

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array notably reduced noise, enhanced image quality, and boosted the contrast.

3.2. In-Sensor Adaptation

When moving from a bright space into darkness, vision initially fails but gradually recovers sensitivity through scotopic adaptation (Figure 4e, top).[47] Conversely, entering a bright space from a dim environment produces temporary blindness, followed by adjustment to high illumination, known as a photopic adaptation (Figure 4e, bottom). These processes are mediated by rod and cone cells, with rods responsible for dim-light detection and cones operating at high intensity and capturing color.[48] The switchover between rods and cones underlies the remarkable adaptability of the human vision.

Inspired by these mechanisms, Cao et al. developed an InP quantum dot and InSnZnO (ITZO) hybrid phototransistor system for in-sensor adaptability (Figure 4f, left). InP quantum dots, owing to their narrow bandgap, strongly absorb visible light (Figure 4f, right). Device measurements revealed that under positive gate bias the drain current (I_{DS}) initially increases and then decreases with time, reflecting an inhibitory response analogous to cone cells. In contrast, under negative gate bias, I_{DS} first decreases but subsequently rises, reproducing the potentiation behavior of rod cells. These behaviors are attributed to carrier trapping and detrapping by defects, highlighting the essential role of gate control in regulating the channel conductance of ITZO transistors (Figure 4g). Figure 4h presents the time-dependent I_{DS} response under continuous illumination, where positive bias induced photopic-like behavior, while negative bias produced scotopic-like characteristics. Using a 4 × 4 device array, these adaptive responses were visualized through the gradual clarification of the patterned images, as shown in Figure 4i. These results demonstrate that metal-oxide semiconductor-based synaptic phototransistors can reproduce biological adaptation phenomena, offering a promising pathway to in-sensor photopic and scotopic adaptation under visible illumination.

3.3. In-Sensor Edge Detection

Edge detection represents a fundamental preprocessing step in vision systems, where redundant background information is suppressed, while preserving sharp transitions. To enable this function in hardware, synaptic devices must exhibit tunable potentiation and depression, allowing convolution-like operations with programmable kernels.[12,49] The ability to realize both excitatory and inhibitory weights within a single platform is particularly critical for implementing filters such as Sobel or Laplacian operators. Hwang et al. addressed this requirement by introducing an artificial optoelectronic synapse composed of two serially connected IGZO phototransistors functioning as excitatory synaptic phototransistor (ESP) and inhibitory synaptic phototransistor (ISP), respectively, as shown in Figure 5a.[50] The device architecture and array scheme are listed in Figure 5b. Under 405 nm optical pulses, irradiation of the ESP channel induced potentiation, whereas irradiation of the ISP produced depression, thereby enabling fully optical bidirectional weight modulation without additional comparators, as shown in Figure 5c. Additionally, modulating the gate voltage allows the operating range to be aligned with the intended value, as shown in Figure 5d. For image preprocessing, each 3 × 3 patch of the input image was converted into a 9 × 1 voltage vector and applied to a synapse array (Figure 5e). The synaptic currents were programmed by optical pulses to emulate kernel weights, with stabilized values after 100 s being used for convolution operations (Figure 5f). Using this approach, sharpening, blurring, Sobel_x and Sobel_y edge detection, and Gaussian blurring were successfully reproduced (Figure 5g). These demonstrations highlight that metal-oxide semiconductor-based optoelectronic synaptic phototransistors not only mimic key synaptic plasticity features but also enable the fully optical bidirectional modulation of synaptic weights with simple circuit integration for edge detection.

3.4. In-Sensor Image Recognition

Image recognition in noisy environments requires devices capable of suppressing background interference while enhancing meaningful features at the sensor level. Kim et al. demonstrated a hybrid architecture integrating IGZO-based optoelectronic synaptic transistors with quantum dot light-emitting diodes (QLEDs) to achieve both noise reduction and direct visualization of ultraviolet (UV) patterns.[51] Figure 5h illustrates an individual IGZO-based optoelectronic synaptic transistors and QLEDs pixel. The optoelectronic synaptic transistors convert noisy UV signals to weighted photocurrents, which are transferred to QLEDs via a current-to-voltage converter. The key difference in current between IGZO optoelectronic synaptic transistors and QLED is that the current of QLED increases much more rapidly (exponential increase) than of IGZO optoelectronic synaptic transistors (linear increase) (Figure 5i). Leveraging the threshold-switching characteristics of QLEDs, suprathreshold signals are amplified into bright visible outputs, while subthreshold noise is suppressed. This dual role enables simultaneous preprocessing and visualization of UV images, markedly reducing background noise, as shown in Figure 5j. When applied to noisy MNIST-derived images, the system produced high-contrast preprocessed outputs that improved recognition accuracy to over 86% in a deep neural network compared with below 50% for conventional IGZO optoelectronic synaptic transistors alone, as shown in Figure 5k and 5l. These results highlight the potential of metal-oxide semiconductor-based optoelectronic synaptic phototransistors, coupled with optoelectronic integration, to perform efficient, real-time, and low-power in-sensor image recognition.

4. CONCLUSION AND PERSPECTIVE

Metal-oxide semiconductor-based optoelectronic synaptic transistors have emerged as promising building blocks for neuromorphic vision, offering intrinsic memory-like behavior through PPC. To overcome the wide bandgap limitation that confines pristine metal-oxide semiconductors to ultraviolet sensitivity, two representative strategies have been demonstrated. Trap-assisted photoexcitation introduces subgap absorption pathways through intentional defect engineering. This strategy not only broadens spectral sensitivity but also amplifies PPC, reinforcing the temporal integration of optical signals. Complementarily, absorption layer engineering integrates low-band gap semiconductors, conjugated polymers, or quantum dots with metal-oxide semiconductor channels. These hybrid structures extend the spectral response into the visible and NIR while simultaneously enabling sustained photocurrent through heterojunction-mediated carrier transfer and photogating. Together, these approaches convert inherent material limitations into functional advantages, establishing metal-oxides as versatile platforms for broadband neuromorphic vision systems.

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Beyond device-level mechanisms, array-level demonstrations highlight the potential of metal-oxide semiconductor-based optoelectronic synaptic transistors in performing essential vision tasks. Contrast enhancement, visual adaptation under changing illumination, edge detection, and even image recognition have been achieved directly within the sensor. These results underline that metal-oxide semiconductor-based synaptic devices are not merely physical demonstrations of PPC-driven dynamics but are capable of executing vision preprocessing steps that reduce data transfer, suppress noise, and enhance efficiency in system-level computing architectures.

The broader trajectory of computing further contextualizes the role of these devices. Figure 6 illustrates the progression

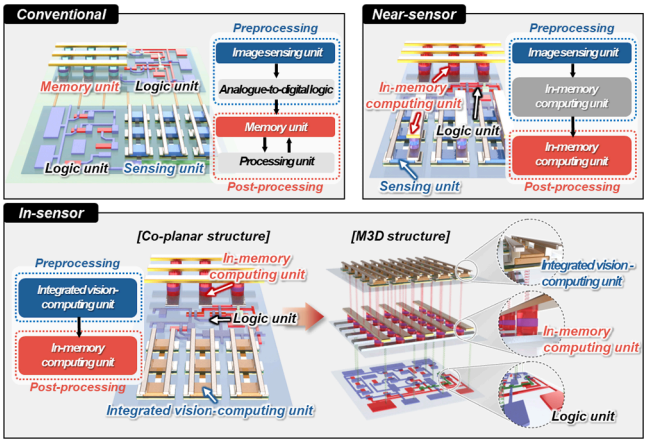


Figure 6. Evolution of sensory-computing architectures toward future in-sensor neuromorphic vision systems, the ultimate direction for metal-oxide semiconductor-based optoelectronic synaptic transistors. Reproduced with permission from ref 52. Copyright 2025 Elsevier.

from conventional von Neumann systems, where sensing, memory, and processing units are isolated, to near-sensor computing, which integrates preprocessing units adjacent to sensors, and ultimately to in-sensor computing, where sensing and computing are colocated.[52] The most advanced form of this paradigm involves M3D integration, stacking vision-sensing, in-memory computing, and vertically stacked logic units in a compact footprint. Such architectures emulate the biological retina–cortex pathway, where preprocessing occurs directly at the point of acquisition, reducing latency, bandwidth demand, and energy consumption. In-sensor neuromorphic vision systems therefore represent the logical end-point of this architectural evolution, combining efficiency, parallelism, and compactness. This M3D integration approach can be practically realized through sequential low-temperature fabrication, in which sensing, memory, and logic tiers are vertically stacked using back-end-of-line (BEOL)-compatible processes.[52] Metal-oxide semiconductors are particularly well suited for such tier-by-tier stacking as key device layers can be deposited and patterned at temperatures below 400 °C, thereby minimizing thermal damage to underlying circuits. At the same time, high-density, large-area sensor arrays with millions of pixels will be required to rival the resolution and performance of commercial imaging platforms, yet maintaining uniform device-to-device performance across such vast arrays is nontrivial.

For all of its promise, M3D integration also introduces additional fabrication and design challenges. Efficient tier partitioning and robust intertier interconnect design are required to prevent routing congestion, excessive signal delay, and unwanted electrostatic coupling as device density increases.[59−61] In addition, the relatively low thermal conductivity of metal-oxide semiconductor materials raises concerns about heat accumulation in densely stacked structures, necessitating advanced thermal management strategies. Reliability remains another critical concern as defect-mediated optoelectronic responses must remain stable over prolonged operation and devices must maintain uniform performance across large-area arrays over their lifetimes. Ultimately, addressing these fabrication, architectural, and reliability challenges will be essential for translating laboratory-scale demonstrations of in-sensor neuromorphic vision systems into practical and scalable M3D-integrated hardware.

System-level challenges also remain. High-density, large-area arrays with millions of pixels are needed to rival commercial imaging platforms. Integration with peripheral circuits, including logic and memory, requires careful codesign to preserve the benefits of in-sensor computing. M3D integration offers the most promising route. By stacking metal-oxide semiconductor-based synaptic layers with logic and memory units, compact and massively parallel sensor–computing arrays can be realized. Such systems would be capable of performing not only preprocessing tasks like contrast enhancement or edge detection but also higher-level recognition and inference directly within the imaging array, enabling edge devices to achieve real-time intelligence without relying on cloud resources.

In conclusion, metal-oxide semiconductor-based optoelectronic synaptic transistors demonstrate how material properties recognized as intrinsic challenges, such as wide bandgaps and defect sensitivity, can be transformed into functional enablers for neuromorphic vision. Through trap-assisted and absorption layer engineering, these devices achieve broadband sensitivity and synaptic plasticity. Through array-level demonstrations, they perform vision preprocessing tasks that mirror retinal functions. Through compatibility with M3D integration, they offer potential pathways toward neuromorphic vision systems. Metal-oxide semiconductors thus stand at the intersection of materials science, device engineering, and neuromorphic computing, uniquely positioned to drive the next generation of intelligent vision systems.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/accountsmr.5c00279.

Basic functionalities of metal-oxide semiconductor-based optoelectronic synaptic transistor (PDF)

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